

A Counterexample to Conjecture 00000008558

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The conjectured equality between Boolean sublattice embedding dimension and the minimum number of lattice generators is false, even for the three-element distributive chain

$$C_3 = \{0 < a < 1\}.$$

Its embedding dimension is 2. Its minimum number of generators is 3 when the lattice operations are $\wedge, \vee$, and is 1 when the bounds 0, 1 are also available as constants. Thus both standard signature conventions contradict the proposed equality.

Embedding dimension is exactly two. For a nonnegative integer n , write $B_n = \mathcal{P}(\{0, \dots, n-1\})$, with intersection and union as meet and join. A sublattice embedding is an injective map preserving both operations. Define

$$f(0) = \emptyset, \quad f(a) = \{0\}, \quad f(1) = \{0, 1\}.$$

These are three distinct, nested subsets. On a chain, meet and join are minimum and maximum; on these nested subsets, intersection and union are the corresponding smaller and larger subsets. Therefore $f : C_3 \hookrightarrow B_2$ is a sublattice embedding. It also preserves bottom and top, so the construction works if that additional convention is imposed.

The lattices B_0 and B_1 have respectively one and two elements, so neither admits an injection from C_3 . Consequently the minimum embedding dimension exists and equals 2. This counterexample lies entirely within the class of lattices which do embed into a finite Boolean lattice.

Generators with the two binary lattice operations. The usual lattice signature has precisely $\wedge$ and $\vee$; the bounded-lattice signature additionally includes two constants [1]. First use the ordinary lattice signature. For any $x, y \in C_3$, both $x \wedge y$ and $x \vee y$ belong to $\{x, y\}$. Induction on lattice terms therefore shows that every term whose inputs lie in a set $S \subseteq C_3$ evaluates to an element already in S . Hence the generated set is exactly S . To generate C_3 , all three elements must be chosen, and the full set does generate it. The minimum number is thus $3 \neq 2$. Generation here means closure under both lattice operations, as in the source's statement about generators of a lattice.

The convention with bounds as constants. If 0 and 1 may be used without being counted as generators, the same term induction shows that S generates exactly $S \cup \{0, 1\}$. A generating set must contain a , and $\{a\}$ does generate the whole chain. The minimum number is therefore $1 \neq 2$. This covers the other standard convention without changing the embedding dimension.

Formal verification and semantic map. The accompanying Lean 4.19.0 file imports only `Std`. The three elements are represented by `Fin 3`, with numerical labels 0, 1, 2 for the bottom, middle, and top respectively. The order, minimum, and maximum satisfy all lattice laws and both distributive identities.

`BooleanLattice n` is the full type of Boolean membership functions on `Fin n`. Intersection and union are pointwise Boolean conjunction and disjunction. Arbitrary proposition-valued

subsets have a proved membership-function encoding. `BooleanEmbedding` requires injectivity and preservation of both operations. The file constructs the two-dimensional embedding and proves that it also preserves bounds. It proves that no embedding in dimensions 0 or 1 exists, quantifying over all maps, rather than merely checking a chosen list of candidate embeddings. `boolean_dimension_two` establishes existence and minimality; `boolean_dimension_unique` identifies every minimum as 2.

Every possible generating set, including every proposition-valued subset, is encoded by its three membership bits. The predicates `Generated` and `BoundedGenerated` are inductively defined by the actual lattice operations, with constants only in the latter. Their characterization is proved by induction, so it covers terms of arbitrary depth. The least-closure universal property is also proved for ordinary generation. The two minimum-generator predicates assert both attainment and a lower bound over every generating set. Their minima are formally proved to be 3 and 1 respectively.

The final theorems

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conjecture_8558_false
conjecture_8558_bounded_signature_false
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negate the existence of a common minimum for dimension and generator number under the two conventions. All three minima have separately been proved to exist, so the disagreement is between actual finite numbers. This disproves the second conjunct of the source conjecture; the separate proposed bound involving join-irreducible elements is not needed. Neither theorem uses an admitted proof, a native proof oracle, or an additional axiom. Both final theorems depend only on `propext` and `Quot.sound`.

References

- [1] S. Burris and H. P. Sankappanavar, *A Course in Universal Algebra*, Millennium Edition, 2012 update, Chapter II, Section 1, examples (8)–(9), and Section 3. <https://www.math.uwaterloo.ca/~snburris/htdocs/UALG/univ-algebra2012.pdf>